The documentation and process conversion measures necessary to comply with this document shall be completed by 7 June 2023.

INCH-POUND

MIL-PRF-19500/453H
w/AMENDMENT 2
7 February 2023
SUPERSEDING
MIL-PRF-19500/453H
w/AMENDMENT 1
4 June 2018

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, HIGH-FREQUENCY, AMPLIFIER
TYPE 2N5109 QUALITY LEVELS JAN, JANTX, JANTXV, JANS, JANHC, JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, VHF-UHF amplifier transistors. Four levels of product assurance (JAN, JANTX, JANTXV and JANS) are provided for each encapsulated device type as specified in MIL-PRF-19500 and two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device type. Provisions for radiation hardness assurance (RHA) to two radiation levels ("R" and "F") are provided for JANTXV product assurance level. Provisions for RHA to eight radiation levels is provided for JANS, JANHC and JANKC product assurance levels. RHA level designators "M", "D", "P", "L", "R", "F", "G", and "H" are appended to the device prefix to identify devices, which have passed RHA requirements.

1.2 Package outlines. The device packages for the encapsulated device types are as follows: (2N5109) (TO-39) in accordance with figure 1, (2N5109UB) (UB) in accordance with figure 2. The dimensions and topography for JANHC and JANKC unencapsulated die is as follows: The A version die in accordance with and figure 3.

1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

P_T (1) T_A = +25°C	P_T (2) T_C = +25°C	V_{CBO}	V_{CEO}	I_C	V_{EBO}	T_{STG} and T_J	$R_{\theta JA}$	$R_{\theta JC}$
W	W	V dc	V dc	A dc	V dc	°C	°C/W	°C/W
1	2.9	40	20	0.4	3.0	-65 to +200	175	60

(1) Derate at 5.71 mW/°C above T_A > +25°C.
(2) Derate at 16.6 mW/°C above T_C > +25°C.

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1.4 Primary electrical characteristics (common to all types).

Limits	h_{FE} V_{CE} = 15 V dc I_C = 50 mA dc	$V_{CE(SAT)}$ I_C = 100mA dc I_B = 10 mA dc	C_{obo} I_E = 0 100 kHz ≤ f ≤ 1 MHz V_{CB} = 28 V dc	$\lvert h_{FE} \rvert$ V_{CE} = 15 V dc I_C = 50 mA dc f = 200 MHz	Power gain I_C = 50 mA dc f = 200 MHz Pin = -10dB V_{CE} = 15 V dc
		V dc	pF		dB
Min	40			6.0	11.0
Max	150	0.5	3.5	11.0	

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level designators.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", “JANTXV”, and "JANS".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest for JANS quality levels are as follows: "M", "D", "P", "L", "R", "F", "G", and "H" For the RHA levels for TXV quality levels are as follows: "R" and "F".

1.5.3 Device type. The designation system for the device types covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the semiconductors covered by this specification sheet are as follows: "5109".

1.5.4 Suffix symbols. The following suffix letters are incorporated in the PIN for this specification sheet.

	A blank first suffix symbol indicates a metal can package (TO-39) in accordance with figure 1.
UB	Indicates a surface mount package in accordance with figure 2.

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QML 19500.

1.5.6 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet is "A".

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1.6 Radiation features. The following radiation features are applicable for RHA devices supplied to this specification sheet.

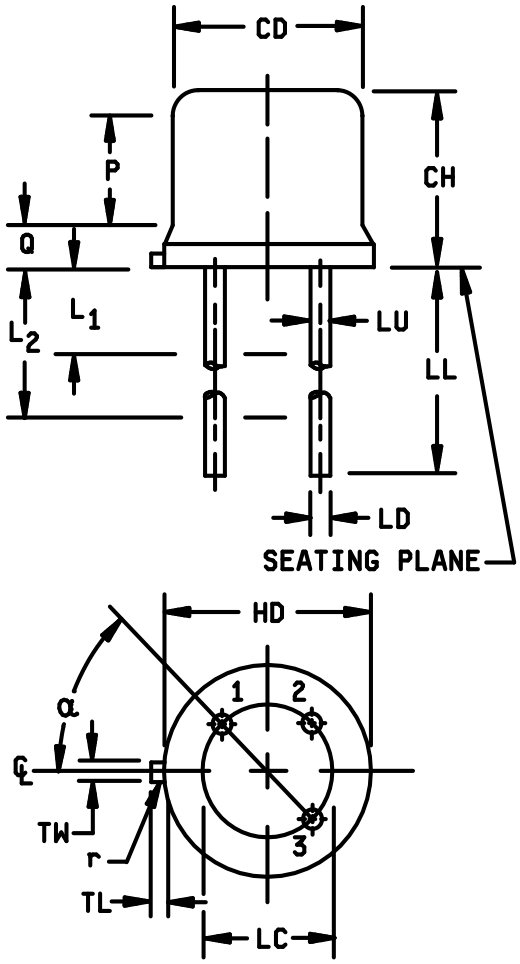
1.6.1 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750 are as follows:

For device type 2N5109, 2N5109UB: ..... 300 krads(Si) 1/

1/ The manufacturers supplying these device types have performed characterization testing in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750. The radiation end point limits are guaranteed only to a maximum TID level of 300 krads(Si).

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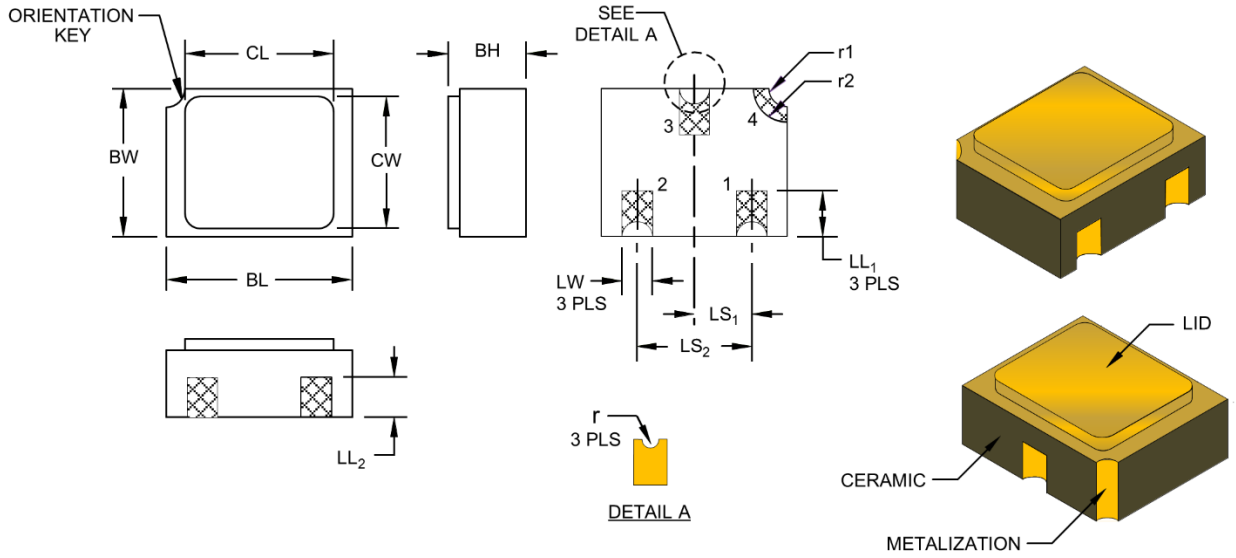
Symbol	Dimensions				Notes
	in	in	mm	mm	
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.190	.210	4.83	5.33	
LD	.016	.021	0.41	0.53	4
LL	.500	.750	12.70	19.05	4
LU	.016	.019	0.41	0.48	4
L_1		.050		1.27	4
L_2	.250		6.35		4
Q		.050		1.27	6
TL	.029	.045	0.74	1.14	5
TW	.028	.034	0.71	0.86	
r		.010		0.25	
α	45° TP		45° TP		
P	.100		2.54		3



NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Lead 1 is emitter, lead 2 is base, and lead 3 is collector.
3. This zone is controlled for automatic handling. The variation in actual diameter within this zone shall not exceed .010 inch (0.254 mm).
4. (Three leads) LU applies between L_1 and L_2. LD applies between L_2 and .5 inch (12.70 mm) from seating plane. Diameter is uncontrolled in L_1 and beyond .5 inch (12.70 mm) from seating plane.
5. Measured from maximum diameter of the actual device.
6. Details of outline in this zone optional.

FIGURE 1. Physical dimensions of transistor type 2N5109 (similar to TO-39).

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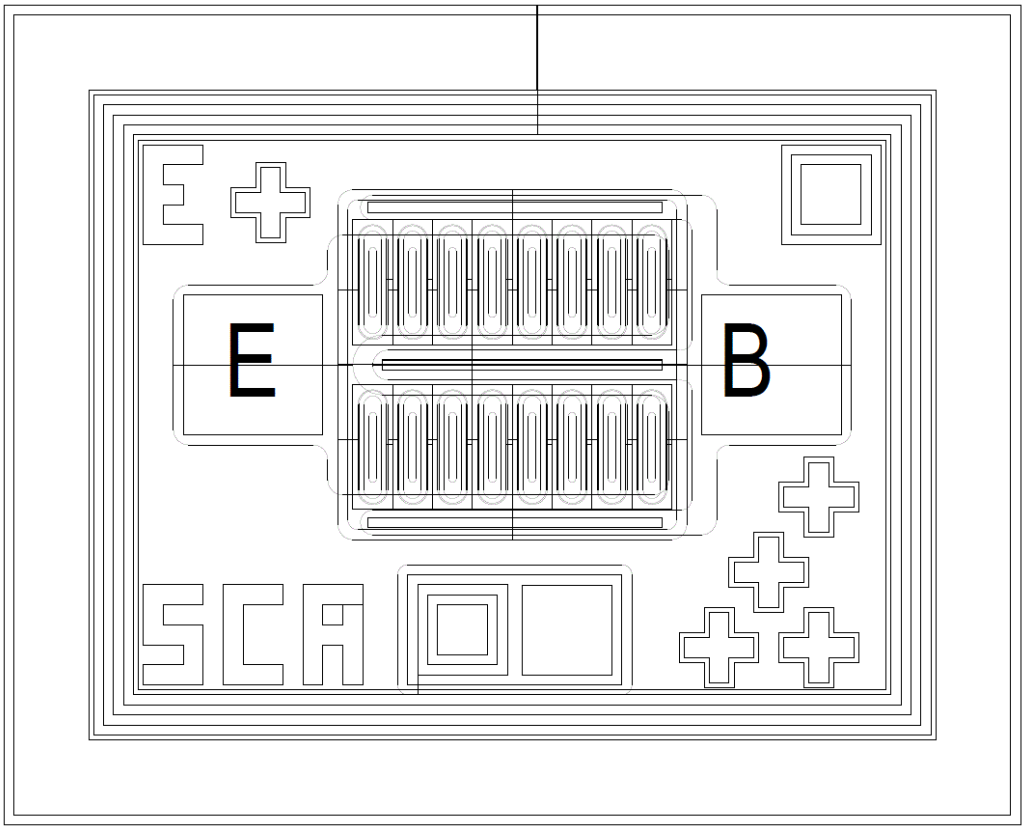


Symbol	Dimensions				Symbol	Dimensions			
	Inches	Inches	mm	mm		Inches	Inches	mm	mm
	Min	Max	Min	Max		Min	Max	Min	Max
BH	.046	.056	1.17	1.42	LS_1	.035	.039	0.89	0.99
BL	.115	.128	2.92	3.25	LS_2	.071	.079	1.80	2.01
BW	.085	.108	2.16	2.74	LW	.016	.024	0.41	0.61
CL		.128		3.25	r		.008		0.20
CW		.108		2.74	r1		.012		0.31
LL_1	.022	.038	0.56	0.97	r2		.022		0.56
LL_2	.017	.035	0.43	0.89					

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.
3. Hatched areas on package denote metallized areas.
4. Lid material: Kovar.

FIGURE 2. Physical dimensions, surface mount UB package (2N5109UB).

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A-version

NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Unless otherwise specified tolerance is .00100 inch (0.0254 mm).
3. Physical characteristics for A-version die are:
   - Die size: .016 x .020 inches (0.4064 x 0.508 mm).
   - Die thickness: .008 ±.0016 inches (0.2032 ±0.04064 mm).
   - Base pad: .0028 x .0028 inches (0.07112 x 0.07112 mm).
   - Emitter pad: .0028 x .0028 inches (0.07112 x 0.07112 mm).
   - Collector: Back side of die.
4. The metallization characteristics of the die are:
   - Back metal: Gold, 5,000 ±1,000 Å
   - Top metal: Aluminum, 17500 ±2500 Å
   - Glassivation: SiO_2, 6,000 ±1,000 Å

FIGURE 3. Physical dimensions and characteristics for JANHC and JANKC (A-version) die.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

cm ..... Cross modulation
G ..... Voltage gain.
NF ..... Noise figure

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (similar to TO-39), figure 2 (2N5109UB) and figure 3 (JANHC2N5109 and JANKC2N5109) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Pin-out. The pin-out of the device shall be as shown on figures 1, 2, and 3 as applicable.

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3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5.1 Surface mount packages. The marking on the UB and UBM packages shall consist of an abbreviated part number, the date code, and the manufacturer's symbol or logo. The prefixes JAN, JANTX, JANTXV and JANS can be abbreviated as J, JX, JV, and JS respectively. The "2N" prefix and the "UB" suffix can also be omitted.

3.5.2 RHA identifiers. The RHA designator shall immediately precede (or replace) the device "2N" identifier (depending upon degree of abbreviation required).

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 RHA. RHA requirements and test levels shall be as defined in MIL-PRF-19500.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

d. Element evaluation of un-encapsulated die (see 4.6).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.3 Radiation hardened devices. See 4.4.4 herein and MIL-PRF-19500.

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4.3 Screening (quality levels JANS, JANTX and JANTXV only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANS levels	Measurements for JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9	I_{CEO1} and h_{FE1}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CEO1} and h_{FE1}; ΔI_{CEO1} = 100 percent of initial value or 2 µA dc, whichever is greater. Δh_{FE1} = ±20 percent.	I_{CEO1} and h_{FE1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CEO1} = 100 percent of initial value or 2 µA dc, whichever is greater. Δh_{FE1} = 20 percent change from initial reading.	Subgroup 2 of table I herein; ΔI_{CEO1} = 100 percent of initial value or 2 µA dc, whichever is greater. Δh_{FE1} = 20 percent change from initial reading.

(1) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc; T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. Power shall be applied to the device to achieve a minimum P_D = 100 percent of P_T maximum rated as defined in 1.3 herein. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for JAN, JANTX, and JANTXV. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW}, (and V_H where appropriate). See table III, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with 4.4.2.1 for quality level JANS and 4.4.2.2 for quality levels JAN, JANTX, and JANTXV.

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4.4.2.1 Quality level JANS. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIA (JANS) of MIL-PRF-19500 and herein.

Subgroup	Method	Conditions
B4	1037	V_{CB} = 10 to 30 V dc, adjust power or current to achieve a ΔT_J = +100°C.
B5	1027	V_{CB} = 10 to 30 V dc; P_D = 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with table E-VIA of MIL-PRF-19500, adjust T_A to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0, adjusted T_A to achieve a T_J = +225°C minimum.

4.4.2.2 Quality levels JAN, JANTX and JANTXV. Group B inspection for quality levels JAN, JANTX, and JANTXV shall be conducted in accordance with the conditions specified in table E-VIC (small die flow) of MIL-PRF-19500 and herein.

Step	Method	Conditions
1	1026	Steady-state life: 1,000 hours, V_{CB} = 10 to 30 V dc; power shall be applied and ambient adjusted to achieve T_J = +150°C using a minimum power dissipation of $P_D \geq$ 100 percent of max rated P_T as defined in 1.3 herein. n = 45 devices, c = 0. The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

4.4.2.2.1 Group B sample selection. Samples selected for small die flow group B inspection shall be in accordance with all of the following requirements:

a. Samples shall be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection.

b. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (step 1) may be pulled prior to the application of final lead finish.

c. Separate samples may be used for each step.

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4.4.2.2.2 Failures. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during conformance inspection shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot and/or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failures mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX and JANTXV) herein for group C testing.

4.4.3.1 Group C inspection, table E-VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, (not applicable to 2N5109UB).
C5	3131	$R_{\theta JC}$ = 60°C/W.
C6	1026	1,000 hours at V_{CB} = 10 - 30 V dc; power shall be applied and ambient temperature shall be adjusted to achieve T_J = +150°C minimum and a power dissipation of P_D ≥ 100 percent of max rated P_T as defined in 1.3 herein shall be applied. n = 45 devices, c = 0. The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

4.4.3.2 Group C inspection, table E-VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E (not applicable to 2N5109UB).
C5	3131	$R_{\theta JC}$ = 60°C/W.
C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS and JANKC types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750, for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2.e herein), except group D, subgroup 2 may be performed separate from other subgroups. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

4.4.5 Group E inspection. Group E inspection shall be performed in accordance with table III herein for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification that did not request the performance of table III tests, the tests specified in table III herein must be performed to maintain qualification.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.6 Element evaluation of unencapsulated die (JANHC and JANKC). The element evaluation of JANHC and JANKC unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical 3/ examination	2071					
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Temperature cycling (air to air) 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal 4/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250C at t = 24 hrs or T_A = +300°C at t = 2 hrs, n = 11 wires, c = 0				
Subgroup 2						
Thermal impedance 6/	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 100 µA dc	$V_{(BR)CBO}$	40		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 5 mA dc	$V_{(BR)CEO}$	20		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_C = 100 µA dc	$V_{(BR)EBO}$	3.0		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 5 mA dc; R_2 = 10 Ω	$V_{(BR)CER}$	40		V dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 15 V dc	I_{CE01}		20	µA dc
Forward-current transfer ratio	3076	V_{CE} = 15 V dc; I_C = 50 mA dc; pulsed (see 4.5.1)	h_{FE1}	40	150	
Collector to emitter saturated voltage	3071	I_C = 100 mA dc; I_B = 10 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.5	V dc
Subgroup 3						
High temperature operation:		T_C = +175°C				
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 15 V dc	I_{CE02}		5.0	mA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 3 – Continued						
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 50 mA dc; pulsed (see 4.5.1)	h_{FE2}	15		
Subgroup 4						
Open circuit output capacitance	3236	V_{CB} = 28 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		3.5	pF
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 15 V dc; f = 200 MHz	$\lvert h_{fe} \rvert$			
		I_C = 25 mA dc; I_C = 50 mA dc; I_C = 100 mA dc		5.0 6.0 5.0	10 11.0 10.5	
Power gain (narrow band) current		V_{CC} = 15 V dc; I_C = 50 mA dc; f = 200 MHz; pin = -10 dB; (see figure 4)	G_{PE}	11		dB
Cross modulation		V_{CC} = 15 V dc; I_C = 50 mA dc; 54 dB output; (see figure 4)	cm		-57	dB
Noise figure		V_{CC} = 15 V dc; I_C = 10 mA dc; f = 200 MHz; pin = -10 dB; (see figure 4)	NF		3.5	dB
Voltage gain (wideband)		V_{CC} = 15 V dc; I_C = 50 mA dc; f = 50 to 216 MHz; pin = -10 dB (see figure 5)	G	11		dB
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, unless otherwise specified see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ For end-point measurements, this test is required for the following subgroups:
Group B, step 1 of 4.4.2.2 herein (JAN, JANTX, and JANTXV).
Group B, subgroups 3, 4, and 5 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1 and 2.

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TABLE II. Group D inspection and end-point limits.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1 4/						
Neutron irradiation	1017	Neutron exposure, V_{CES} = 0 V				
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 100 µA dc	$V_{(BR)CBO}$	40		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 5 mA dc	$V_{(BR)CEO}$	20		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_C = 100 µA dc	$V_{(BR)EBO}$	3.0		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 5 mA dc; R_2 = 10 Ω	$V_{(BR)CER}$	40		V dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 15 V dc	I_{CEO1}		40	µA dc
Forward-current transfer ratio	3076	V_{CE} = 15 V dc, I_C = 50 mA dc; pulsed (see 4.5.1)	$[h_{FE1}]$	5/ [20]	150	
Collector to emitter saturated voltage	3071	I_C = 100 mA dc; I_B = 10 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.58	V dc
Subgroup 2						
Steady-state total dose irradiation	1019	Condition A, Gamma exposure V_{CES} = 16 V				
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 100 µA dc	$V_{(BR)CBO}$	40		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 5 mA dc	$V_{(BR)CEO}$	20		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_C = 100 µA dc	$V_{(BR)EBO}$	3.0		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 5 mA dc; R_2 = 10 Ω	$V_{(BR)CER}$	40		V dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 15 V dc	I_{CEO1}		40	µA dc
Forward-current transfer ratio	3076	V_{CE} = 15 V dc, I_C = 50 mA dc; pulsed (see 4.5.1)	$[h_{FE1}]$	5/ [20]	150	
Collector to emitter saturated voltage	3071	I_C = 100 mA dc; I_B = 10 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.58	V dc

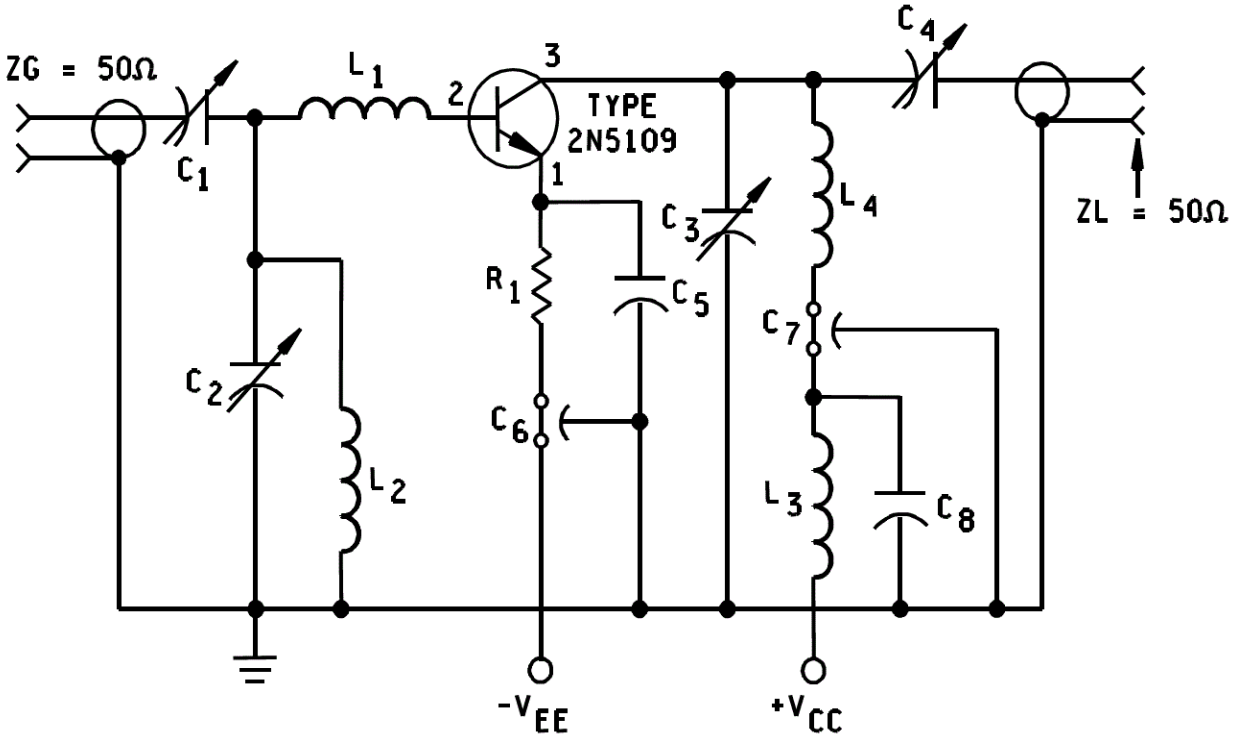
1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to all device types unless otherwise noted.
4/ See 6.2.e herein.
5/ See method 1019 of MIL-STD-750, for how to determine $[h_{FE}]$ by first calculating the delta ($1/h_{FE}$) from the pre and post-radiation h_{FE}. Notice that $[h_{FE}]$ is not the same as h_{FE} and cannot be measured directly. The $[h_{FE}]$ value can never exceed the pre-radiation minimum h_{FE} that it is based upon.

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TABLE III. Group E inspection (all quality levels) – for qualification or re-qualification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal	1071	Fine and gross leak.	
Electrical measurements		See table I, subgroup 2.	
Subgroup 2			45 devices, c = 0
Intermittent life	1037	V_{CB} = 10 V dc, 6000 cycles, adjust power or current to achieve a ΔT_J = +100°C.	
Electrical measurements		See table I, subgroup 2.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			
Not applicable			
Subgroup 6			
ESD	1020	Testing is required for nonsensitive listing to prove capability.	
Subgroup 8			45 devices, c = 0
Reverse stability	1033	Condition B.	

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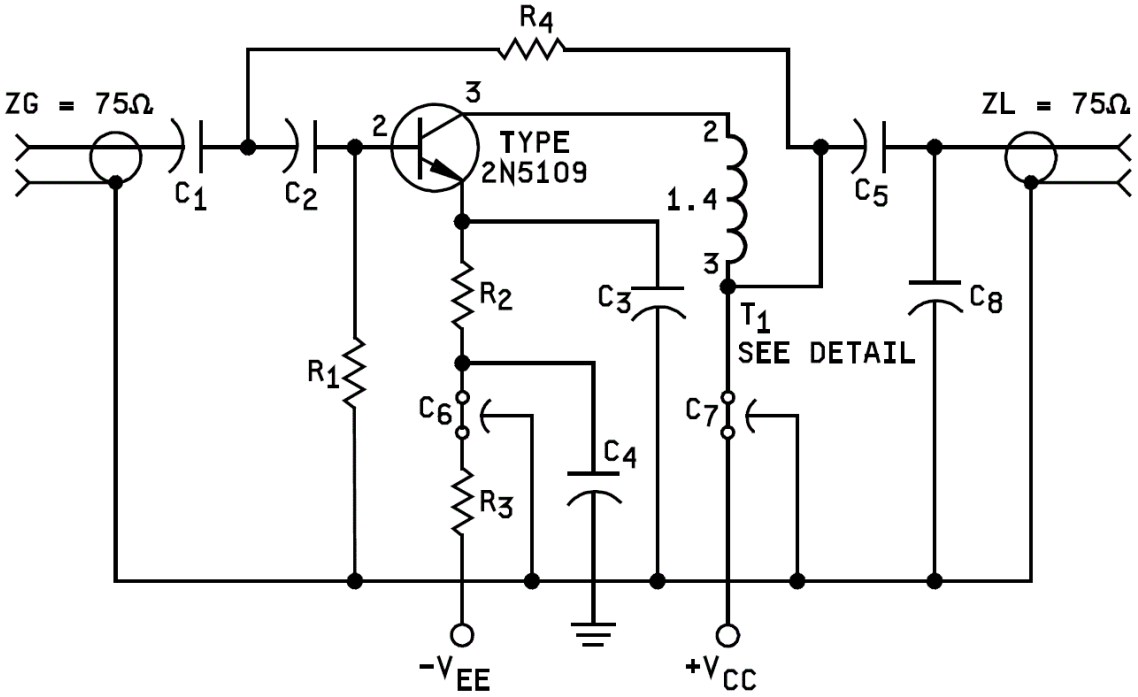


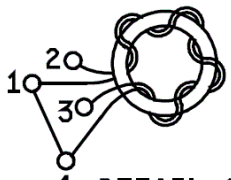
C_1, C_2, C_3: 1.0 to 30 pF.
C_4: 1.0 to 20 pF.
C_5: 10,000 pF
C6, C7: 1,000 pF.
C8: .01 µF.

L1 4.5 turns, number 22 wire, .187 inch I. D.
L4 3.5 turns, number 22 wire, .187 inch I. D.
L2, L3 0.82 µH RFC.
R1 240 Ω, 2 watts.

FIGURE 4. RF amplifier for power gain and noise figure tests.

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DETAIL OF TRANSFORMER T_1

C_1, C_2, C_3, C_5: 0.002 µF.
C_4: 0.03 µF.
C_6, C_7: 1500 pF.
C_8: 18 pF.
R_1: 4.7 kΩ, 0.25 W.
R_2: 6.8 Ω, 0.5 W.
R_3: 330 Ω, 1W.
R_4: 200 Ω, 0.25 W.
T_1: 4 turns number 30 wire.

Bifilar wound on "Indiana General".
Core number CF-102-Q1, or equivalent.

FIGURE 5. RF amplifier for voltage gain test.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

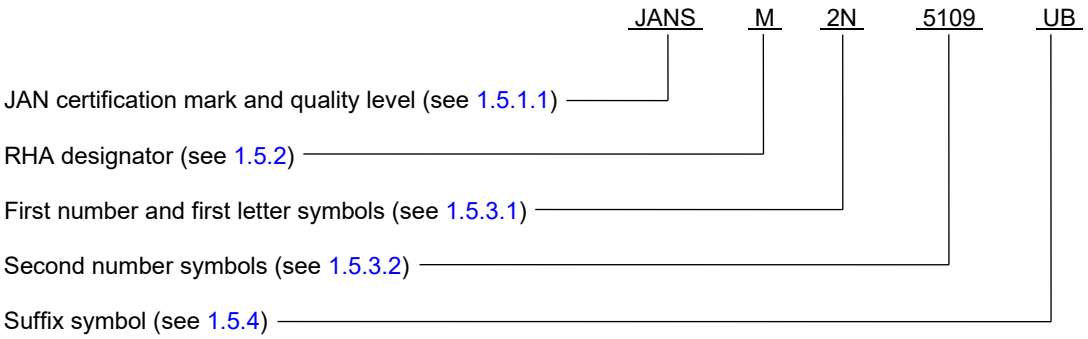
d. The complete PIN, see 1.5 and 6.5.

e. For acquisition of RHA designed devices, table II, subgroup 1 testing of group D is optional. If subgroup 1 testing is desired, it should be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

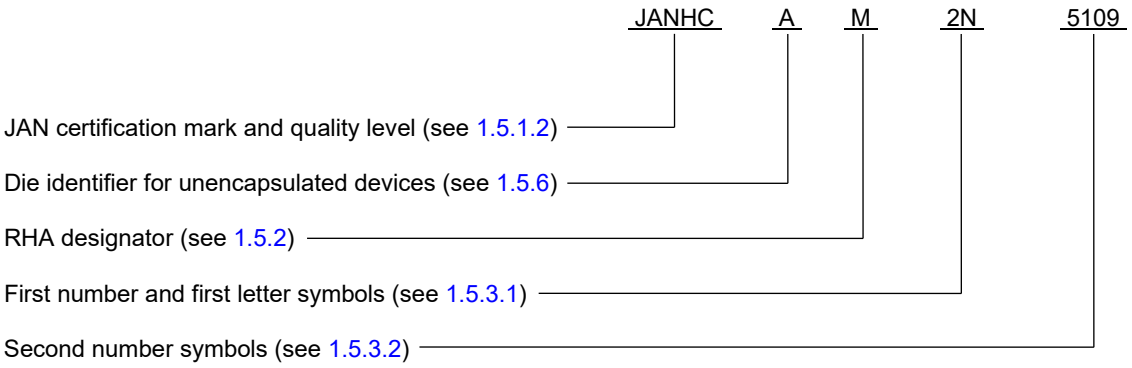
6.4 PIN construction examples.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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6.4.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

PINs for encapsulated 2N5109 1/	
JAN2N5109	JAN2N5109UB
JANTX2N5109	JANTX2N5109UB
JANTXV2N5109	JANTXV2N5109UB
JANTXVR2N5109	JANTXVR2N5109UB
JANTXVF2N5109	JANTXVF2N5109UB
JANS#2N5109	JANS#2N5109UB
PINs for un-encapsulated JANC die 1/	
JANHCA#2N5109	JANKCA#2N5109

(1) The number sign (#) represent one of eight RHA designators available (M, D, P, L, R, F, G, or H). The PIN is also available without a RHA designator.

6.6 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N5109) will be identified on the QML.

Die ordering information	
PIN	Manufacturer
	34156
2N5109	JANHCA2N5109 JANKCA2N5109

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6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693-1642 or DSN 850-6939.

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Air Force - 85
DLA - CC
NASA - NA

Review activity:
Army - MI
Air Force - 19, 71

Preparing activity:
DLA - CC

(Project 5961-2023-014)

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